

WGB-GLFI: A Novel Graph-Based Global-Local Feature Interaction Framework for Automated Seizure Detection

Xiang Li[#], Mingxing Zhu[#], Chuqi Yang[#], Ke Zhang, Xin Wang, Sunday Timothy Aboyeji, Fei Chen, Chen Yao, Xiaomao Fan, Xianhai Zeng, Yan Chen, Changmiao Wang* and Shixiong Chen*

Abstract—Epilepsy detection faces significant challenges due to unpredictable seizures, ranging from brief awareness lapses to severe convulsions, posing risks to patients' safety and quality of life. In recent years, deep

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Xiang Li, Mingxing Zhu and Chuqi Yang contributed equally.

Xiang Li, Chuqi Yang and Shixiong Chen is with the School of Medicine, The Chinese University of Hong Kong, Shenzhen, Guangdong 518172, China (e-mail: 224050069@link.cuhk.edu.cn).

Mingxing Zhu is with the School of Electronics and Information Engineering, Harbin Institute of Technology, Shenzhen, Guangdong 518055, China (e-mail: bestarpc@126.com).

Ke Zhang is with the School of Medicine, The Chinese University of Hong Kong, Shenzhen, Guangdong 518172, China, and the Department of Otolaryngology, Shenzhen Children's Hospital, Shenzhen, Guangdong 518000, China (e-mail: kezhang@cug.edu.cn).

Xin Wang is with the Guangdong-Hong Kong-Macao Joint Laboratory of Human-Machine Intelligence-Synergy Systems, Shenzhen Institutes of Advanced Technology, Chinese Academy of Sciences, Shenzhen, Guangdong 518055, China (e-mail: wangxin@siat.ac.cn).

S. T. Aboyeji is with the Department of Otolaryngology, Shenzhen Longgang Otolaryngology Hospital, and also with Shenzhen Otolaryngology Research Institute, Shenzhen, Guangdong 518172, China and the School of Medicine, The Chinese University of Hong Kong, Shenzhen, Guangdong 518172, China. (e-mail: sundayaboyej@ieee.org).

Fei Chen is with the Research Institute of Tsinghua University in Shenzhen and Shenzhen Eartech Co., Ltd, Shenzhen, Guangdong 518063, China (e-mail: chenfei@tsinghua-sz.org).

Chen Yao is with the Department of Neurosurgery, The National Key Clinic Specialty, Shenzhen Key Laboratory of Neurosurgery, the First Affiliated Hospital of Shenzhen University, Shenzhen Second People's Hospital, Shenzhen, Guangdong 518035, China (e-mail: pro.yaochen@gmail.com).

Xiaomao Fan is with the College of Big Data and Internet, Shenzhen Technology University, Shenzhen, Guangdong 518118, China (e-mail: fanxiaomao@sztu.edu.cn).

Xianhai Zeng is with the Department of Otolaryngology, Shenzhen Longgang Otolaryngology Hospital, Shenzhen, Guangdong 518172, China, and Shenzhen Otolaryngology Research Institute, Shenzhen, Guangdong 518172, China (e-mail: zxhklwx@163.com).

Yan Chen is with the Surgery Division, Epilepsy Center, Shenzhen Children's Hospital, Shenzhen, Guangdong 518036, China (e-mail: ccflysz@126.com).

Changmiao Wang is with the Shenzhen Research Institute of Big Data, Shenzhen, Guangdong 518038, China (e-mail: cmwangalbert@gmail.com).

Corresponding authors: Changmiao Wang (e-mail: cmwangalbert@gmail.com), Shixiong Chen (e-mail: chenshixiong@cuhk.edu.cn).

learning has become a mainstream approach in this field, leveraging advanced computational resources and EEG datasets. However, a key challenge remains: existing methods often lack unified spatial modeling and struggle to effectively handle local detailed features, thereby limiting their accuracy and robustness. To address these issues, we propose the Weighted Graph Building Global-Local Feature Interaction (WGB-GLFI) framework, which integrates spatial connectivity and dynamic patterns through a Weighted Graph Building (WGB) module and a Global-Local Feature Interaction (GLFI) module. This approach excels by comprehensively capturing the dynamic spatial relationships during epileptic seizures and achieving seamless global-local feature integration, significantly enhancing seizure detection performance. Its effectiveness has been validated across multiple datasets, including CHB-MIT, Siena Scalp, and private datasets, demonstrating robust and reliable results. Evaluated on these datasets, our model achieves accuracy rates of 99.28%, 99.21%, and 99.30%, respectively. The reliability and robustness of our framework provide epilepsy patients with faster and more reliable seizure detection, which helps to intervene in a timely manner and improve the quality of life of patients.

Index Terms—Seizure Detection, Spatiotemporal Framework, Weighted Graph, Global-Local Interaction

I. INTRODUCTION

EPILEPSY is a chronic neurological disorder characterized by recurrent, unprovoked seizures caused by abnormal electrical activity in the brain [1]. It affects approximately 50 million people worldwide, making it one of the most common neurological conditions globally [2]. Seizures can vary in type and severity, ranging from brief lapses in awareness to prolonged convulsions, and their unpredictability poses serious risks to patients' safety and quality of life. In clinical practice, accurate and timely detection of epileptic seizures is crucial for effective disease management, treatment planning, and preventing potentially life-threatening consequences such as status epilepticus or Sudden Unexpected Death in Epilepsy (SUDEP) [3]. However, manual seizure annotation from long-term electroencephalogram (EEG) recordings is time-consuming and requires expert knowledge, motivating the development of automated seizure detection methods based on machine learning and deep neural networks [4].

With the increasing availability of EEG datasets and advances in computational resources, deep learning-based ap-

proaches have become the mainstream solution for automatic seizure detection [5] [6]. These methods excel at modeling the non-linear and non-stationary nature of EEG signals, enabling end-to-end learning frameworks that bypass traditional feature engineering. Temporal models like Long Short-Term Memory (LSTM) [7] [8] and Temporal Convolutional Networks (TCNs) [9] [10] effectively capture sequential dependencies in EEG time-series data. Attention mechanisms and graph-based models like Graph Attention Networks (GATs) [11] effectively model inter-channel relationships and brain spatial dynamics. EEG temporal patterns, from short-term variations to long-term trends, are captured by models handling local and global dependencies. Recent research has focused on hybrid architectures that integrate global and local spatio-temporal features to better capture the characteristics of epileptic seizures for improved detection. For instance, Zhou et al. [12] developed the Global-Local Neural Epilepsy Detection Network (GlepNet), a model that interweaves temporal convolutional layers with multi-head attention mechanisms in the GlepNet encoder block to jointly extract local and global seizure-related features from EEG signals. Similarly, Zhao et al. [13] proposed an approach for interactive local and global feature coupling in EEG-based epileptic seizure detection, utilizing convolutional operations for local feature extraction and self-attention mechanisms for global feature modeling. In many cases, such frameworks rely on shallow fusion strategies and lack explicit mechanisms to enable rich, bidirectional information exchange across different feature levels. As a result, these methods often fail to fully leverage the interactions between global and local semantic information. Moreover, enabling mutual learning between low-level and high-level semantic features could significantly enhance the model's expressiveness and improve seizure detection performance [14].

Researchers have explored the spatio-temporal characteristics of EEG signals for seizure detection, yet effectively integrating these characteristics remains challenging, particularly due to limitations in spatial information modeling. The spatio-temporal nature of EEG requires models to jointly capture spatial connectivity and dynamic patterns in order to represent the complex manifestations of epileptic seizures [15]. However, spatial information, which reflects functional connectivity and interactions between brain regions, is often either ignored or modeled only at a coarse level in conventional architectures [16]. Many existing methods rely on fixed electrode positions or static distance matrices to represent spatial relationships, which cannot capture critical abnormal patterns such as dynamic inter-regional synchronization. Others separate the modeling of temporal and spatial features or adopt shallow fusion strategies, making it difficult to effectively learn the nonlinear interactions between the two modalities. Furthermore, these methods generally lack hierarchical modeling mechanisms for multi-scale temporal patterns, which reduces the ability to capture both short-term bursts and long-range correlations, and they fail to fully exploit the contextual relationships between global brain activity and localized abnormal patterns [17], [18]. These limitations restrict both the accuracy and robustness of seizure detection, highlighting the need for a framework that jointly models spatial relationships and multi-

scale information to extract comprehensive contextual features and discriminative patterns from EEG signals.

To address the limitations of existing approaches in jointly modeling spatial and temporal dependencies, we propose a unified spatiotemporal framework for seizure detection. Our model consists of two core components: a weighted graph building module to capture spatial relationships and a feature interaction module to integrate dynamic patterns, enhanced by a knowledge distillation module utilizing KL divergence, which enables low-level and high-level semantic information to dynamically learn from each other during the model updating process, thereby effectively improving the model's performance. By fusing global and local representations, our framework effectively captures the complex temporal structure of epileptic EEG signals, leading to improved robustness and generalization across subjects.

The main contributions of the study are as follows:

- **WGB-GLFI Framework:** We propose a novel spatiotemporal framework, Weighted Graph Building Global-Local Feature Interaction (WGB-GLFI), for epileptic seizure detection using EEG signals. This architecture integrates spatial connectivity and dynamic temporal patterns to address the limitations of existing methods in capturing complex seizure-related features, achieving high accuracy and robustness across diverse datasets.
- **Spatial Information Fusion and Global-Local Interaction:** This module integrates a Weighted Graph Building approach, using a correlation matrix and TCN to capture inter-channel dependencies and temporal features within each 1-second EEG window, and fuses them via GAT [11] for enhanced spatial information integration. Meanwhile, the Global-Local Feature Interaction module employs Shifted Causal Attention (SCA) and Residual Convolutional Network (RCN) blocks to model multi-scale temporal patterns, enhanced by bidirectional KL divergence loss distillation for robust global-local feature interaction, effectively addressing challenges in spatial modeling and feature fusion.
- **Extensive Experiments:** We evaluate the proposed model on three datasets: CHB-MIT Dataset, Siena Scalp EEG Dataset, and Private Dataset. The experimental results show that our method consistently outperforms several state-of-the-art baseline methods in terms of Accuracy, Sensitivity, F1, and Kappa, demonstrating the effectiveness of the proposed spatiotemporal fusion architecture.

The remainder of this paper is organized as follows: Section II presents the proposed WGB-GLFI framework and its core modules, Section III describes the datasets, preprocessing, evaluation metrics, and experimental setup, Section IV provides the results from 5-fold and leave-one-patient-out cross-validation strategies, Section V discusses ablation experiments and module validity, and Section VI concludes the study.

II. METHODS

Our seizure detection framework is illustrated in Fig. 1. Firstly, the seizure detection framework denoises EEG data

and then preprocesses it with Fast Independent Component Analysis (FastICA). Secondly, it processes the data through a Weighted Graph Building module and a Global-Local Feature Interaction module, and aligns local and global features using KL divergence loss to guide RCNs in learning globally-aware local features through distillation. Finally, it concatenates the outputs for comprehensive final classification.

A. Weighted Graph Building

To capture individual seizure patterns, a dynamic graph structure is built where nodes represent EEG channels and edges reflect their spatial interdependencies. This framework effectively models spatial changes during seizures, enhancing adaptability to individual differences and improving seizure pattern recognition.

1) *Channel adjacency matrix*: To represent the spatial relationships between EEG signal channels, a $N \times N$ connectivity matrix is generated using Pearson correlation, where N denotes the total number of EEG channels. Each element C_{ij} represents the correlation between channels i and j . The matrix is symmetric, with diagonal elements $C_{ii} = 1$ (self-correlation) and off-diagonal elements showing correlations between different channels.

The correlation is calculated using the following formula:

$$C_{ij} = \frac{\text{Cov}(R_i, R_j)}{\sigma_{R_i} \sigma_{R_j}}, \quad (1)$$

where:

- $\text{Cov}(R_i, R_j)$ represents the covariance between the i -th and j -th channels;
- σ_{R_i} and σ_{R_j} represent the standard deviations of the i -th and j -th channel signals, respectively.

The C_{ij} obtained from this formula is the Pearson correlation coefficient, which reflects the spatial correlation between the EEG channels. By using Pearson correlation, the spatial seizure patterns can be effectively captured, enhancing the analysis of dynamic brain activity during epileptic events [19].

2) *Channel Feature matrix*: To capture the temporal features within each time window, as shown in Fig. 2, we generate a feature matrix for each channel using Temporal Convolutional Network (TCN), which reflects the temporal dynamics of each channel. The output feature matrix for each channel has the dimensions $N \times D$, where N is the number of channels and D is the feature dimension, typically determined by the kernel size and the number of layers. To ensure the causality of the temporal data, TCN uses causal convolutions, ensuring that the output at the current time step only depends on the current and previous time steps, avoiding the influence of future information, thus maintaining the sequential integrity of the data, which is especially suitable for tasks such as seizure detection.

The dilated convolution is mathematically represented as:

$$y(t) = \sum_{i=0}^{k-1} f(i) \cdot x(t - d \cdot i), \quad (2)$$

where $y(t)$ is the output of the convolution, $f(i)$ is the i -th element of the convolution kernel, $x(t - d \cdot i)$ is the dilated input

signal, d is the dilation factor controlling the spacing between elements of the kernel, and k is the size of the convolution kernel.

By using TCN, we can expand the receptive field without increasing the computational burden, thus capturing longer-range temporal features and enhancing the model's ability to handle long-term dependencies in the data [20]. These features are then combined with those from other channels to construct a multi-dimensional representation of the entire EEG signal.

3) *Dependency Modeling of GAT in Channels*: In this module, we apply the Graph Attention Network (GAT) to capture the relationships between EEG signal channels. By assigning attention coefficients to the edges of the graph, GAT allows the model to adaptively focus on the most relevant channels, thereby improving seizure detection accuracy [21]. The input to GAT is a graph where each node represents an EEG channel, and the edges represent the relationships between channels. These relationships are constructed by the feature matrices generated by the Pearson correlation and TCN modules along the feature dimension, which are then further processed by GAT to enhance the representation of channel interactions. To further improve performance, we select the Top- k % most influential edges based on the attention coefficients, discarding the remaining $1 - k$ edges. The input dimensions are: the channel adjacency matrix is $N \times N$, the channel feature matrix is $N \times D$, and the final output is $N \times d$.

The mathematical formulation for a given node i is as follows:

$$h'_i = \sum_{j \in \mathcal{N}(i)} \alpha_{ij} W h_j, \quad (3)$$

where h'_i is the updated feature representation of node i , $\mathcal{N}(i)$ represents the set of neighboring nodes of node i , α_{ij} is the attention coefficient between node i and node j , W is a learnable weight matrix, and h_j is the feature representation of node j .

The attention coefficient α_{ij} is computed using the following mechanism:

$$\alpha_{ij} = \frac{\exp(\text{LeakyReLU}(a^T [W h_i || W h_j]))}{\sum_{k \in \mathcal{N}(i)} \exp(\text{LeakyReLU}(a^T [W h_i || W h_k]))}, \quad (4)$$

where:

- a is a learnable attention vector;
- LeakyReLU is the activation function used to compute the attention scores.
- $||$ denotes the vector concatenation operator.

GAT performs graph convolution by calculating attention coefficients to dynamically weight and aggregate neighboring node features, learned based on node features and their relationships. This enables effective capture of complex EEG channel dependencies, enhancing model focus on relevant channels and improving analysis performance. The attention mechanism adaptively learns the importance of neighboring channels for better detection.

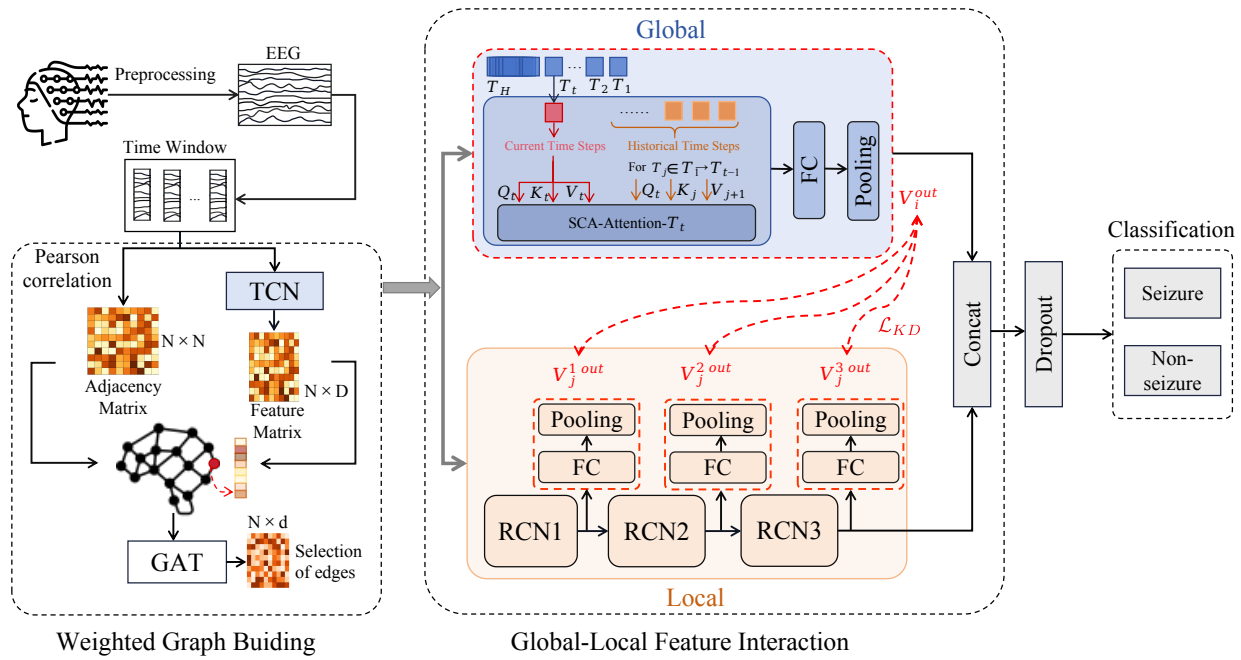


Fig. 1: Framework of the Proposed WGB-GLFI Model. The framework includes Weighted Graph Building and Global-Local Feature Interaction. First, data flows from EEG preprocessing to the Weighted Graph Building module, capturing spatial relationships. Second, it proceeds to the Global-Local Feature Interaction module, culminating in enhanced seizure detection through integrated spatial and temporal analysis. Finally, it classifies the data into seizure and non-seizure categories.

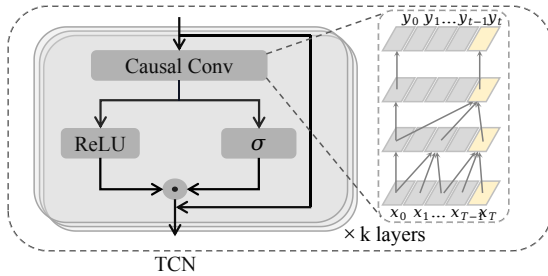


Fig. 2: The structure of the TCN consists of dilated convolutions, ReLU activation, and a final output layer, designed to capture temporal dependencies across multiple stacked layers.

B. Global Local Feature Interaction

This module aims to enhance EEG signal analysis for seizure detection by integrating global and local feature modules. Global features capture long-term inter-channel dependencies, while local features target short-term temporal dynamics. Global-local interaction improves the fusion of both information types.

1) SCA-Attention Based Global Feature Extraction Module:

This module introduces Shifted Causal Attention (SCA) to enhance global feature extraction in EEG signals, especially in capturing long-range dependencies across channels [22]. Moreover, SCA preserves causality while enabling the model to aggregate information more effectively across the temporal dimension. Here, $t \in [H]$, meaning that the sequence contains a total of H time steps. Given the query, key, and value vectors q_t, k_t, v_t for $t \in [H]$, we define the output of SCA as:

$$SCA_t = \frac{\sum_{j=1}^{t-1} \exp(\langle q_t, k_j \rangle) v_{j+1} + \exp(\langle q_t, k_t \rangle) v_t}{\sum_{j=1}^t \exp(\langle q_t, k_j \rangle)}. \quad (5)$$

The core idea of SCA is that, at time step t , the model attends to all previous keys k_j ($j < t$). For each past position j the associated value used in the attention is the shifted value v_{j+1} , while for the current position t the model uses the unshifted value v_t . This design is reflected in the formula above and preserves strict causality because no information from step $t+1$ is accessed, which prevents any leakage of information about the current target and enables more effective temporal aggregation. In seizure detection, this improved temporal modeling leads to higher detection accuracy and better overall performance.

2) RCN Based Local Feature Extraction Module:

As shown in Fig. 3, this module introduces a local feature extraction method based on RCN, combining the residual connection mechanism of ResNet with the powerful feature extraction capability of Convolutional Neural Networks (CNN), enabling efficient extraction of local features from EEG signals and capturing fine-grained temporal dependencies and spatial patterns in the EEG signals. Note that the RCN module consists of five Conv1D branches with kernel sizes 5, 7, 9, 11 and 13 (each with 32 output channels), where a notation such as “Conv1D 5x32” denotes a 1D convolution using a kernel size of 5 and 32 channels. This design allows the model to extract local features from EEG signals at multiple scales, ensuring the identification of signal variations across different time scales. The feature maps extracted from all convolution layers are merged through concatenation into a unified feature

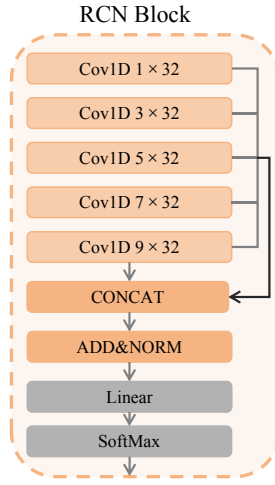


Fig. 3: The RCN Block structure includes a multi-scale feature extraction process using 1D convolutions, with concatenation and softmax for capturing local seizure features.

representation, which enhances the model's ability to capture complex signal patterns.

The RCN module incorporates residual connections and normalization techniques to mitigate issues like vanishing and exploding gradients. The use of residual connections allows information to be effectively propagated through deeper network layers, preventing information loss and improving the optimization of the training process, thereby enhancing the model's generalization ability. In the task of seizure detection, the RCN module efficiently extracts key local features from EEG signals, and through the multi-scale convolution operations, it helps the model capture important temporal patterns before and after a seizure. This allows RCN to not only excel in local feature extraction but also improve the model's learning of temporal dependencies, thereby enhancing its ability to recognize complex patterns such as those found in dynamic EEG signals for seizure detection.

3) Global local feature extraction method based on KL divergence distillation: To comprehensively learn the relationship between global and local features, we adopt the KL divergence distillation method. KL-divergence distillation is used to align the high-level global features with the fine-grained local representations. Specifically, the KL divergence is defined as:

$$D_{KL}(P \parallel Q) = \sum_x P(x) \log \left(\frac{P(x)}{Q(x)} \right), \quad (6)$$

where $P(x)$ represents the probability distribution of global features, and $Q(x)$ represents the probability distribution of local features. By minimizing the KL divergence, the learning process of local features can approximate the behavior of global features, thus optimizing the extraction ability of local features.

The KL divergence distillation occurs between the global feature extraction module and local feature extraction modules. Through KL divergence distillation, the local feature extraction modules gradually approximate the output distribution of the global feature extraction module during training, enhancing

feature extraction accuracy and overall performance.

We define the KL divergence loss function for distillation as follows:

$$\mathcal{L}_{KD} = \frac{1}{N} \sum_{i=1}^N \sum_{j=1}^3 (f_{KL}(V_i^{out}, V_j^{iout}) + f_{KL}(V_j^{iout}, V_i^{out})) \quad (7)$$

where V_i^{out} denotes the output of the global feature extraction module, V_j^{iout} denotes the output of the local feature extraction modules, and $f_{KL}(\cdot)$ measures the difference between global and local features. Here, N is the total number of samples. The index j in the inner summation represents the three local feature extraction modules for each sample i , and $\mathcal{L}_{KD}$ is the KL divergence loss function for distillation between global and local features.

The total loss function is defined as $\mathcal{L} = \mathcal{L}_{CE} + \mathcal{L}_{KD}$, where $\mathcal{L}_{CE}$ is the binary cross-entropy loss for epilepsy detection and $\mathcal{L}_{KD}$ is the KL divergence loss. Minimizing KL divergence enables the local feature extraction modules to gradually align with the global feature extraction module, improving feature extraction accuracy. This cooperative approach optimizes both local feature representation and global feature aggregation, leading to better performance in tasks like epilepsy seizure detection.

III. EXPERIMENTS

A. Datasets

1) CHB-MIT Dataset: This study uses the CHB-MIT Dataset [23], which contains EEG recordings from 22 patients with refractory epilepsy. The dataset includes 969 hours of EEG data, covering 23 cases, and records 173 seizure events. All data were recorded using a Nihon Kohden EEG-1100A system and sampled at 256 samples per second with a 16-bit resolution. Notably, case chb21 is from the same patient as chb01, but the data were collected 1.5 years later. The dataset includes various types of seizures, such as clonic, atonic, and tonic seizures, with patient ages ranging from 1.5 to 22 years. All EEG recordings use a bipolar montage based on the International 10-20 system, with most files containing 23 EEG signals (24 or 26 in a few cases, due to additional EEG derivations or inclusion of non-EEG signals such as ECG or VNS). In this study, only the EEG channels are used. The redundant T8-P8 channel (which appears duplicated as the last channel in recordings) has been removed [24], as both electrodes of this derivation are already covered by preceding channels in the montage. The selected 22 channels are: FP1-F7, F7-T7, T7-P7, P7-O1, FP1-F3, F3-C3, C3-P3, P3-O1, FP2-F4, F4-C4, C4-P4, P4-O2, FP2-F8, F8-T8, T8-P8, P8-O2, FZ-CZ, CZ-PZ, P7-T7, T7-FT9, FT9-FT10, FT10-T8.

2) Siena Scalp EEG Dataset: The Siena Scalp EEG Dataset [25] consists of EEG recordings from 14 patients acquired at the Unit of Neurology and Neurophysiology of the University of Siena. The dataset includes 9 male subjects (ages 25–71) and 5 female subjects (ages 20–58). The patients were monitored using a Video-EEG system with a sampling rate of

TABLE I: Hyperparameter search ranges and network-based selected configurations.

Hyperparameter	Search Range	CHB-MIT	Siena	Private Dataset
TCN Kernel Size	{3, 5, 7}	7	5	3
GAT Attention Heads	{2, 4, 8}	4	4	2
Hidden Size of Global Block	{64, 128, 256}	128	64	64
Dropout Rate Before Final Classification	{0.1, 0.2, 0.3, 0.5}	0.2	0.5	0.5
Optimizer	{Adam, AdamW}	Adam	Adam	Adam
Learning Rate	{1e-3, 5e-4, 1e-4, 1e-5}	1e-4	1e-4	1e-5
Batch Size	{16, 32, 64}	32	32	32

512 Hz, with electrodes placed according to the international 10-20 system. Most of the recordings also contain 1 or 2 EKG signals. In this study, 19 EEG channels common to all subjects were used: Fp1, Fp2, F3, F4, C3, C4, P3, P4, O1, O2, F7, F8, T3, T4, T5, T6, Fz, Cz, Pz. The data were collected using EB Neuro and Natus Quantum LTM amplifiers, with reusable silver/gold cup electrodes. Patients were instructed to stay in bed as much as possible, either awake or asleep.

3) *Private Dataset*: This private dataset, derived from the Cardiac Intensive Care Unit of Shenzhen Children's Hospital, contains EEG data from infant patients, all of whom are under one year of age. The data was collected between June 2021 and October 2022 using the Nicolet video-EEG system with a sampling rate of 500 Hz. The dataset includes EEG recordings from 12 international 10-20 system electrodes (Fp1, Fp2, C3, C4, T3, T4, P3, P4, O1, O2, Cz, Pz). It covers 20 patients, with an average of 11.15 epileptic seizure events per patient, providing rich and unique data resources for studying the characteristics of epilepsy in infants. Expert neurophysiologists provided time-aligned annotations for the onset and end of epileptic seizures. This retrospective study was approved by the hospital, with informed consent waived, and all data was de-identified. All experimental protocols were approved by the Ethics Committee of Shenzhen Children's Hospital (IRB-20250153).

B. Pre-Processing

For each patient in the dataset, all seizure samples are selected, with non-seizure samples set at 2-4 times the number of seizure samples to increase the amount of training data and reduce the risk of overfitting [26] [27]. A step size of 0.5-second overlap is introduced between time windows to address data imbalance [27]. Moreover, the EEG signals are initially divided into 1-second time windows to capture the temporal dynamics of the brain activity. Within each window, several pre-processing steps are applied sequentially: To reduce artifacts in the EEG acquisition process and improve model efficiency, a band-pass filter ranging from 1 Hz to 40 Hz [28] was first applied to remove unwanted noise while preserving the relevant frequencies for seizure detection. Next, we applied Independent Component Analysis (ICA) using the Fast Independent Component Analysis (FastICA) algorithm to remove independent components (ICs) in the concatenated EEG data. ICA models EEG as a linear mixture of independent sources and is widely used to separate neural activity from artifacts. FastICA was chosen for its high computational efficiency, fast convergence, and robustness in estimating in-

dependent sources [29]. After decomposition, ocular and other non-brain components were removed, with artifact-related ICs confirmed through visual inspection following standard EEG preprocessing practices. Finally, the signals are normalized using z-score standardization to remove individual differences, ensuring uniformity across subjects and preparing the data for further analysis and model training [30].

C. Evaluation Metrics

This experiment evaluates the seizure detection task using the following metrics: Accuracy (AC), Sensitivity (SE), F1 Score (F1), Kappa, and False Positive Rate (FPR).

$$AC = \frac{TP + TN}{TP + TN + FP + FN}, \quad (8)$$

$$SE = \frac{TP}{TP + FN}, \quad (9)$$

$$F1 = 2 \times \frac{\text{Precision} \times \text{Sensitivity}}{\text{Precision} + \text{Sensitivity}}, \quad (10)$$

$$\text{Kappa} = \frac{P_o - P_e}{1 - P_e}, \quad (11)$$

$$\text{FPR} = \frac{FP}{FP + TN}. \quad (12)$$

These metrics have been proven efficient in assessing the performance of the seizure detection algorithms, where TP (True Positive), TN (True Negative), FP (False Positive), and FN (False Negative) represent the different types of classification outcomes, and P_o (observed agreement) and P_e (expected agreement) are used to calculate Kappa, which measures the agreement between the model's predictions and the actual labels, adjusted for chance.

TABLE II: Performance of the 5-Fold Cross-Validation Strategy in Each Dataset.

Dataset	AC (%)	SE (%)	F1 (%)	Kappa (%)	FPR (/h)
CHB-MIT	99.28	99.30	98.56	98.66	0.10
Siena Scalp	99.21	99.36	98.98	98.82	0.14
Private	99.30	99.38	98.63	99.11	0.09

D. Overall Experimental setup

We implemented WGB-GLFI on a high-performance computing environment equipped with an RTX 4090 24GB GPU using PyTorch 2.4.1. The training process spanned 50 epochs and employed an early stopping mechanism that triggered

TABLE III: Comparison of k-fold strategy on the CHB-MIT Dataset with Other State-of-the-Art Methods.

Author	Model	Year	Method	AC (%)	SE (%)	F1 (%)
Zhang et al. [26]	Bi-GRU	2022	10-fold CV	98.49	93.89	-
Shen et al. [31]	DWT+RUSBoosted Tree	2022	5-fold CV	96.38	96.15	-
He et al. [32]	GAT+Bi-LSTM	2022	5-fold CV	98.52	97.75	95.90
Zhao et al. [33]	CNN + Transformer	2023	10-fold CV	98.76	97.70	-
Xu et al. [16]	GCN and BiGRU	2024	5-fold CV	97.35	98.85	97.40
Zhang et al. [34]	DLFE	2024	5-fold CV	98.63	98.06	-
Sun et al. [35]	MCCT-Net	2025	10-fold CV	98.14	97.57	97.85
Gao et al. [36]	Cross-Attention Transformer	2025	10-fold CV	97.03	95.36	-
Dokare et al. [37]	SHAP-RELF	2025	5-fold CV	96.58	94.52	94.72
Fan et al. [38]	PLV+Bi-LSTM-Attention	2025	5-fold CV	98.40	98.94	-
WGB-GLFI (Ours)	WGB-GLFI	2025	5-fold CV	99.28	99.30	98.56

TABLE IV: Comparison of k-fold Strategy on the Siena Scalp EEG Dataset with Other State-of-the-Art Methods.

Author	Model	Year	Method	AC (%)	SE (%)	F1 (%)
Xiong et al. [39]	Improved Genetic Algorithm	2022	5-fold CV	98.88	98.36	98.75
Zhang et al. [26]	Bi-GRU	2022	10-fold CV	96.71	97.44	96.90
Zhao et al. [33]	CNN + Transformer	2023	10-fold CV	97.87	97.40	97.45
Bi et al. [40]	EoE + SE	2024	10-fold CV	-	97.94	98.59
Xu et al. [16]	GCN and BiGRU	2024	5-fold CV	98.11	98.09	96.57
Li et al. [41]	TSFA	2025	10-fold CV	98.89	98.98	98.46
Dokare et al. [37]	SHAP-RELF	2025	10-fold CV	94.81	94.04	94.28
Sun et al. [35]	MCCT-Net	2025	10-fold CV	99.08	98.95	98.02
Deng [42]	Guided-CNN Transformer	2025	5-fold CV	99.18	98.88	98.98
Li et al. [43]	GLWA	2025	5-fold CV	98.22	98.29	97.37
WGB-GLFI (Ours)	WGB-GLFI	2025	5-fold CV	99.21	99.36	98.98

TABLE V: Comparison of k-fold Strategy on the Private Dataset with Other State-of-the-Art Methods.

Author	Model	Year	Method	AC (%)	SE (%)	F1 (%)
Zhang et al. [26]	Bi-GRU	2022	10-fold CV	94.22	96.61	96.99
He et al. [32]	GAT+Bi-LSTM	2022	5-fold CV	98.77	98.23	98.65
Zhao et al. [33]	CNN + Transformer	2023	10-fold CV	96.84	98.24	98.39
Zhang et al. [34]	DLFE	2024	5-fold CV	95.27	98.21	97.76
Xu et al. [16]	GCN and BiGRU	2024	5-fold CV	95.32	98.50	96.11
Zhou et al. [12]	GlepNet	2024	10-fold CV	97.18	98.66	97.27
Fan et al. [38]	PLV+Bi-LSTM-Attention	2025	5-fold CV	98.19	99.04	98.18
Gao et al. [36]	Cross-Attention Transformer	2025	10-fold CV	96.97	96.65	97.07
Sun et al. [35]	MCCT-Net	2025	10-fold CV	98.07	97.98	97.92
Dokare et al. [37]	SHAP-RELF	2025	10-fold CV	96.13	97.99	97.01
WGB-GLFI (Ours)	WGB-GLFI	2025	5-fold CV	99.30	99.38	98.63

if the loss did not decrease for 5 consecutive epochs. The remaining hyperparameters, including those of the TCN, GAT, and Global Block modules, were determined through network-based hyperparameter search, as summarized in Table I, and all training was conducted independently for each dataset without any cross-dataset mixing.

IV. RESULTS

A. Experimental results on the 5-Fold Cross-Validation strategy

To ensure robust model performance evaluation across diverse patient data, we employed standard 5-fold cross-validation on the dataset, where each fold uses 80% of the data for training and 20% for testing. As summarized in Table II, the model achieves average accuracies of 99.28%, 99.21%, and 99.30% on the CHB-MIT, Siena Scalp EEG, and private datasets respectively, demonstrating strong adaptability and stability across variations in data quality, patient heterogeneity, and recording conditions such as electrode types, sampling

frequencies, and recording environments. The results also show consistently high F1 scores and sensitivities, with the highest sensitivity reaching 99.38%, and all datasets maintaining low false positive rates (0.09–0.14%), indicating the model's ability to preserve high detection accuracy while effectively minimizing false alarms. These findings confirm the model's robustness and applicability in diverse clinical scenarios, effectively addressing challenges posed by heterogeneous data and complex recording conditions.

1) *Comparison with other advanced methods:* As shown in Tables III, IV, and V, our comparison with other state-of-the-art methods on the CHB-MIT Dataset, Siena Scalp EEG Dataset, and the Private Dataset using k-fold cross-validation demonstrates that our method, WGB-GLFI, achieves higher accuracy, sensitivity, and F1 score, showcasing superior overall performance. For instance, on the CHB-MIT Dataset, the GCN and BiGRU method of Xu et al. [44] achieves a sensitivity of 98.85% and an F1 score of 97.40%, while the GAT+Bi-LSTM of He et al. [32] reports 98.52% accuracy and 95.90% F1

score, both falling short of WGB-GLFI's performance. Similarly, on the Siena Scalp EEG Dataset, the Improved Genetic Algorithm of Xiong et al. [39] achieves 98.88% accuracy and 98.75% F1 score, while the Guided-CNN Transformer of Deng [42] reaches 99.18% accuracy and 98.98% F1 score, yet their performances are inferior to those in the proposed WGB-GLFI. In addition, although MCCT-Net of Sun et al. [35] achieves 99.08% accuracy, 98.95% sensitivity, and 98.02% specificity, its overall performance remains slightly lower than that of the proposed WGB-GLFI. Moreover, on the private dataset, our framework also surpasses GlepNet of Zhou et al. [12] and MCCT-Net of Sun et al. [35], achieving the highest accuracy and F1 score among all compared methods. By integrating Weighted Graph Building and Global Local Feature Interaction modules, our approaches significantly enhance classification performance, demonstrating notable advantages in seizure detection tasks across both datasets.

TABLE VI: Performance of the Leave-One-Patient-Out Cross-Validation Strategy in Each Dataset.

Dataset	AC (%)	SE (%)	F1 (%)	Kappa (%)	FPR (/h)
CHB-MIT	91.02	91.11	89.97	90.05	0.36
Siena Scalp	90.96	89.99	89.22	89.72	0.42
Private	92.01	92.23	91.24	91.30	0.29

B. Experimental results on the Leave-One-Patient-Out Cross-Validation strategy

To comprehensively assess the performance and generalizability of our model across diverse patient data, we conducted experiments using the Leave-One-Patient-Out Cross-Validation (LOPOCV) strategy. In this approach, for a dataset with n patients, $n-1$ patients are used for training, while the remaining patient is reserved for testing, ensuring that the test patient's data is not involved in the training process. This method maximizes training data utilization and provides an unbiased estimate of model performance on unseen patients. The average results for all patients within each dataset are presented in Table VI. For instance, on the Private Dataset, our model achieves an accuracy of 92.01% and a sensitivity of 92.23%, while on the CHB-MIT Dataset, it records an F1 score of 89.97% and a Kappa of 90.05%. These results demonstrate consistent performance across varied seizure patterns and recording conditions. The LOPOCV approach ensures robust evaluation by testing on unseen patient data, enhancing the model's reliability in real-world clinical scenarios. As shown in Table VII, which presents results under patient-independent evaluation strategies, our model outperforms other state-of-the-art methods on the CHB-MIT Dataset, achieving superior accuracy and sensitivity compared to approaches such as the Attention+Bi-LSTM model by Yao et al. [46], which reported an accuracy of 83.89% and a sensitivity of 83.72% under the Leave-One-Out Cross-Validation (LOOCV) protocol. In addition, our method also surpasses the MGCNA framework proposed by Li et al. [47], which achieved an accuracy of 89.45% and a sensitivity of 84.72% under the Leave-One-Subject-Out Cross-Validation (LOSOCV) setting. Furthermore, although

Zhou et al. [51] proposed both the SE-FCN and TransNet models, which achieved accuracies of 87.00% and 86.00% respectively with corresponding sensitivities of 73.00% and 79.00%, their performances still fall short of the proposed WGB-GLFI.

V. DISCUSSION

A. Ablation study

1) *Effectiveness of Global Local Feature Interaction:* To evaluate the effectiveness of each component in the Global Local Feature Interaction, we conducted a comprehensive ablation study. Fig. 4 presents the module-ablation study results for global-local feature interaction, focusing on accuracy of different modules across various datasets. The Transformer + RCN + KL Divergence model (our WGB-GLFI), achieves the highest performance across all datasets. The RCN model alone yields the lowest scores, e.g., 85.97% on CHB-MIT and 87.13% on Siena Scalp EEG Dataset, indicating its limited capability in capturing complex relational patterns. The Transformer baseline performs better, but combining it with RCN significantly improves results, reaching 96.15% on CHB-MIT Dataset and 97.06% on Siena Scalp EEG Dataset, and KL Divergence further enhances performance by refining feature alignment, achieving up to 99.28% on CHB-MIT Dataset and 99.30% on Private Dataset. We employed the paired t-test to analyze the differences in results from the module ablation experiments. The findings revealed statistically significant variations in accuracy across different module combinations, underscoring the robustness of our integrated approach. These findings emphasize the synergistic effect of integrating Transformer, RCN, and KL Divergence, highlighting the effectiveness of our WGB-GLFI method for optimal detection performance.

2) *The influence of parameter k in Weighted Graph Building:* In the construction of the weighted graph using GAT, different choices of the parameter k result in different brain network structures. Therefore, we select k values from 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90% to investigate the sensitivity of the results to k . This approach allows us to analyze how variations in k affect brain network construction and subsequent task performance, with the aim of determining the most suitable k value to improve the effectiveness and precision of the model.

As depicted in Fig. 5, the impact of Top $k\%$ edges on model accuracy across the CHB-MIT, Siena Scalp EEG, and Private datasets reveals distinct trends across varying edge proportions from 10% to 90%. Specifically, model accuracy increases with channels inclusion up to the Top-60% to Top-70% range, reaches a peak, and then declines. All three datasets exhibit optimal performance within the Top-60% to Top-70% edge inclusion range. Specifically, the CHB-MIT Dataset achieves its highest accuracy of 99.28% at Top-70%, the Siena Scalp EEG Dataset peaks at 99.21% at Top-60%, and the Private Dataset reaches its maximum accuracy of 99.30% at Top-70%. This analysis highlights that model accuracy improves with increasing edge inclusion up to the 60%–70% range, beyond which additional channels introduce redundancy or

TABLE VII: Comparison of CHB-MIT Dataset Performance Using Patient-Independent Evaluation Strategies Against State-of-the-Art Methods.

Author	Model	Year	Method	AC (%)	SE (%)
Duan et al. [45]	Deep Metric Learning	2021	LOOCV	86.68	79.64
Yao et al. [46]	Attention+Bi-LSTM	2021	LOOCV	83.89	83.72
Li et al. [47]	MGCNA	2024	LOSOCV	89.45	84.72
Aboyemi et al. [24]	DCSENet	2025	LOPOCV	87.29	85.36
Mohamed et al. [48]	GT-STAFG	2025	LOPOCV	-	80.70
Ye et al. [49]	CPRSCA-ResNet	2025	LOSOCV	78.71	78.70
Li et al. [43]	GLWA	2025	LOPOCV	89.84	89.52
Shao et al. [50]	FCN + SE	2025	LOPOCV	87.00	79.60
	SE-FCN		LOPOCV	87.00	73.00
Zhou et al. [51]	TransNet	2025	LOPOCV	86.00	79.00
WGB-GLFI (Ours)	WG-GLFI	2025	LOPOCV	91.02	91.11

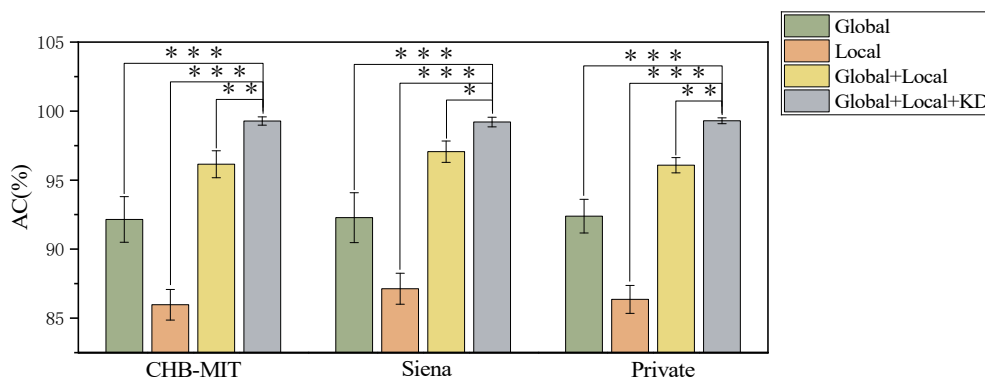


Fig. 4: Ablation Experiments on the Effectiveness of Each Module. Statistical significance is indicated as follows: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

TABLE VIII: Impact of RCN Number on Model Performance.

Dataset	Layer Depth	AC (%)	SE (%)	F1 (%)	Kappa (%)	FPR (/h)
CHB-MIT	1	92.42	93.90	91.60	92.13	0.15
	2	96.08	95.80	95.47	93.37	0.12
	3	98.88	97.65	97.48	98.00	0.12
	4	99.28	99.30	98.56	98.66	0.12
	5	99.28	99.30	98.56	98.66	0.10
Siena Scalp EEG	1	91.70	92.75	92.58	92.99	0.16
	2	95.98	95.80	96.18	96.66	0.17
	3	98.99	98.14	98.24	98.82	0.15
	4	99.21	99.36	98.99	98.82	0.14
	5	99.21	99.36	98.90	98.82	0.14
Private	1	94.50	94.80	94.00	94.20	0.09
	2	98.80	98.90	97.20	96.40	0.10
	3	99.30	99.07	98.38	99.11	0.11
	4	99.30	99.38	98.63	99.11	0.09
	5	99.30	99.38	98.63	99.11	0.09

noise, leading to a decline in performance. These results emphasize the need for a balanced edge selection strategy, tailored to each dataset's characteristics, to achieve optimal detection performance.

3) The Influence of the Number of RCN in Global Local Feature Interaction : To investigate the effectiveness of RCN quantity on model performance, we varied the number of RCNs in our experiments. Table VIII illustrates the impact of varying the number of RCNs on model performance across the CHB-MIT, Siena Scalp EEG, and Private datasets, showing a general trend where performance improves with more RCNs

up to 3 or 4, then stabilizes. For the CHB-MIT Dataset, accuracy increases from 92.42% with 1 RCN to 99.28% at 4 RCNs, where it stabilizes. For the Siena Scalp EEG Dataset, accuracy rises from 91.70% with 1 RCN to 99.21% at 4 RCNs, where it stabilizes. For the Private Dataset, accuracy improves from 94.50% with 1 RCN to 99.30% at 3 RCNs, stabilizing at 4 RCNs, with recall, F1-score, and Kappa peaking at 99.38%, 98.63%, and 99.11%, respectively, at 4 RCNs. These results indicate that performance generally improves with increasing RCNs up to 3 or 4, beyond which additional RCNs yield minimal gains, suggesting an optimal range of 3 to 4 RCNs

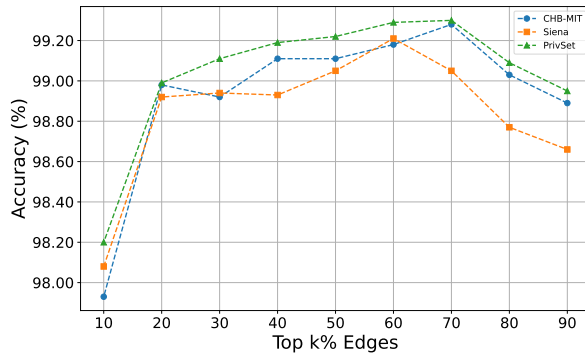


Fig. 5: Ablation Experiments on the Parameter k.

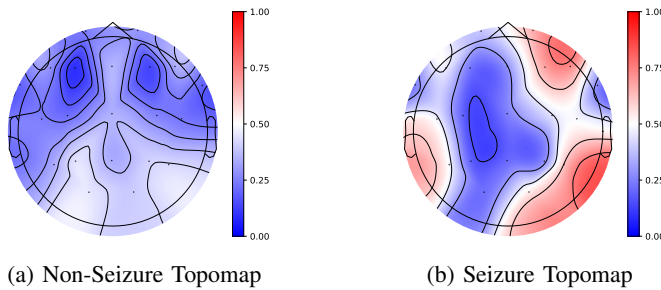


Fig. 6: Comparison of Non-Seizure and Seizure Topomaps Visualized via Weighted Graph Buiding, which can more effectively highlight critical regions by focusing on relevant neural connections.

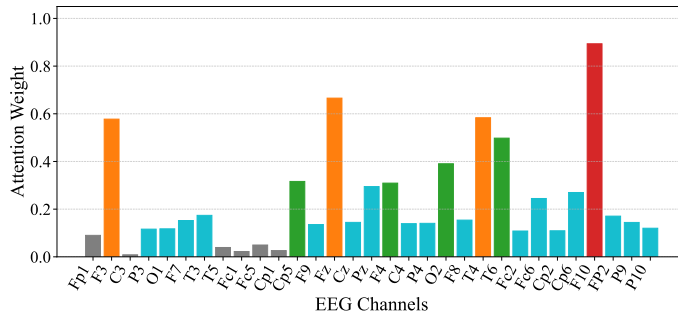


Fig. 7: Visualization of channel weights for the PN10 patient from the Siena Scalp Dataset, employed to assess the contribution levels of different channels during epileptic seizures.

for balancing performance and computational efficiency across all three datasets.

B. The validity of the Temporal-Spatial Module

1) *The effectiveness of GAT in the Weighted Graph Building module:* In this study, we selected two patients for analysis: CHB10 from the CHB-MIT Dataset and PN10 from the Siena Scalp Dataset, to validate the spatial effectiveness of the Weighted Graph Building module. Initially, we visualized the weights of different channels derived from the Weighted Graph Building module, computed the average weight for each channel, and normalized these values to the range [0, 1]. These normalized weights were visualized as an attention

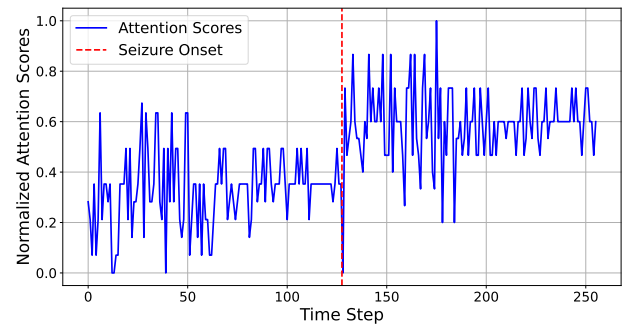


Fig. 8: The SCA-Attention score for a 1-s time window of data exhibits a significant trend, with a seizure onset occurring at the 128th time step.

heatmap based on electrode positions, as depicted in Fig. 6. The heatmap reveals that attention scores are significantly higher post-seizure compared to pre-seizure, demonstrating the model's capability to effectively capture seizure-related spatial patterns. Distinct regions of high attention are evident, particularly in the frontal and temporal areas, suggesting that the model prioritizes these spatially significant regions for enhanced feature interaction during epileptic events.

Transitioning to the analysis in the PN10 patient from the Siena Dataset, we examined the contributions of various EEG channels during seizure events, based on normalized attention weights derived from the Weighted Graph Building module. Fig. 7 illustrates the distribution of attention weights across channels, where Fz and F10 exhibit notably high weights, indicating that the frontal lobe, particularly the right frontal and midline regions, serves as the primary contributing area during epileptic seizures. This finding aligns with the pathological mechanisms of frontal lobe epilepsy. The temporal lobe, including channels such as T4 and T6, is also relatively active, hinting at potential seizure propagation from the frontal lobe to the right temporal lobe, forming a fronto-temporal network. The Weighted Graph Building module enhances network representation by dynamically adjusting node weights, enabling more precise reflection of critical brain region interactions during seizures, thereby improving the reliability and accuracy of lesion identification.

2) *The effectiveness of SCA-Attention in the Weighted Graph Building module:* Fig. 8 illustrates the trend of SCA-Attention scores for a 1-second time window of data from the CHBMIT Dataset, with a seizure onset at the 128th time step. It is evident that during the seizure period, the SCA-Attention score significantly increases, reflecting the model's sensitivity to seizure-related patterns. This rapid response highlights the temporal effectiveness of the SCA-Attention mechanism in capturing critical changes within a short time frame. Furthermore, the ability to detect such shifts in real-time underscores its potential for timely seizure monitoring and intervention.

Moreover, in the Global-Local Feature Interaction (GLFI) module, to investigate the effectiveness of our proposed Shifted Causal Attention (SCA), we conducted ablation experiments by replacing it with other attention mechanisms and evaluated their performance in seizure detection tasks. The

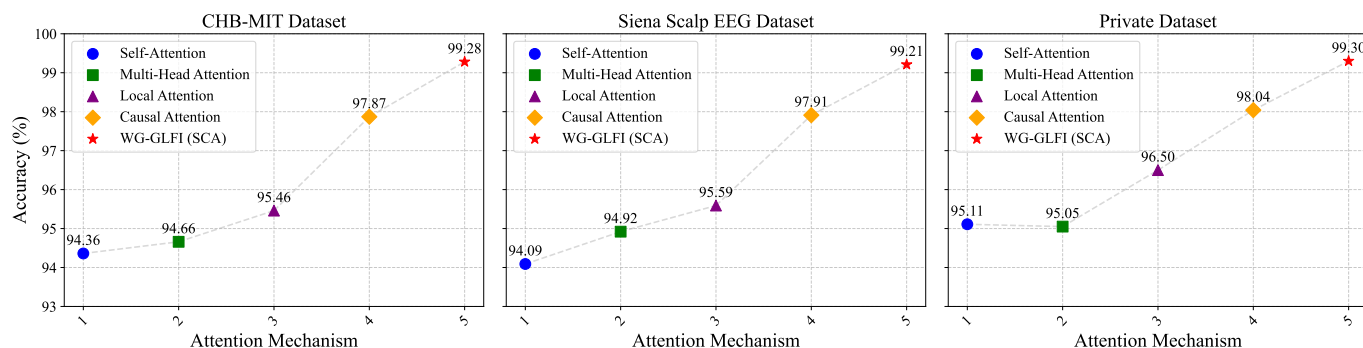


Fig. 9: Accuracy of different attention mechanisms across CHB-MIT, Siena Scalp EEG, and Private Datasets.

tested attention mechanisms are as follows:

- **Self-Attention** [52]: Captures global dependencies in the sequence through dot-product operations on queries, keys, and values, but lacks temporal causality, potentially leading to information redundancy.
- **Multi-Head Attention** [52]: Enhances feature representation by computing multiple attention heads in parallel, but is computationally complex and limited in modeling temporal dynamics.
- **Local Attention** [53]: Restricts attention computation to a fixed time window, suitable for capturing local dependencies, but may overlook long-range feature interactions.
- **Causal Attention** [54]: Ensures attention focuses only on current and previous time steps via a masking mechanism, preserving temporal causality. Unlike SCA, it lacks a time-step shifting mechanism to dynamically integrate future local information for enhanced contextual modeling.

All experiments were conducted with hyperparameters consistent with the settings described in Section III. As shown in Fig. 9, Causal Attention improves performance by incorporating causal constraints, achieving, for instance, 98.04% on the Private dataset, but it still falls short in capturing the complex dynamic features of EEG signals due to the absence of a dynamic shifting mechanism. In contrast, our SCA, by combining causal constraints with time-step shifting, significantly enhances the modeling of global and local features, achieving accuracies of 99.28%, 99.21%, and 99.30% on the CHB-MIT, Siena Scalp EEG, and Private datasets, respectively, demonstrating the best performance trend and proving its superior effectiveness in seizure detection.

VI. CONCLUSION

In this study, we introduced WGB-GLFI, a novel spatiotemporal framework for epileptic seizure detection from EEG signals. WGB-GLFI incorporates a weighted graph building module for spatial relationship modeling with a global-local feature interaction module enhanced by knowledge distillation, effectively capturing the complex dynamic characteristics of EEG data. Comparative studies demonstrate its superior performance and clinical potential, providing a solid foundation for advancing clinical EEG analysis with promising applications in neurological diagnostics. Future plans involve refining

the proposed framework for challenging cases and exploring its utility in other complex neurological conditions.

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